

Victor Lam

+1 (484) 563-5509 | victorlam204@yahoo.com | Pasadena, CA

<https://www.linkedin.com/in/viclam/>

SUMMARY

Seasoned Senior Full-Stack Engineer with over 14 years of experience designing, developing, and scaling high-performance web applications, AI-driven platforms, and data-intensive systems across diverse industries. With deep expertise in Python (Django, FastAPI, Flask), JavaScript/TypeScript (React, Node.js, Next.js), and cloud-native architectures (AWS, GCP, Azure), I bring a unique blend of backend engineering, frontend development, and data science proficiency.

I have successfully led teams and guided startups and enterprises through complex technical challenges—building scalable microservices, optimizing data pipelines, and integrating AI/ML models into production systems. My experience spans data engineering, predictive analytics, and web scraping, using technologies such as PySpark, PostgreSQL, Go, and modern DevOps tooling to deliver reliable, data-driven solutions.

EXPERIENCE

Senior Software Engineer

Jul 2020 - Present

Spokeo

- Led a cross-functional engineering team to design, develop, and maintain an enterprise-grade People Search Engine leveraging React/Next.js, D3.js, Node.js, Python, and FastAPI.
- Architected and scaled distributed search platforms using OpenSearch, designing high-performance query pipelines for person and property intelligence using Python, Go, and PostgreSQL.
- Integrated AI-powered microservices with OpenAI and Google Gemini APIs to interpret natural language queries, improving backend translation accuracy and search relevance across millions of requests.
- Enhanced system reliability and SEO performance by optimizing high-traffic properties, achieving and maintaining near-perfect Lighthouse scores under production load.
- Spearheaded modernization of the company's core platform using Next.js, Tailwind CSS, Go, Python, and Docker, improving scalability, maintainability, and developer velocity.
- Drove the technical implementation for Spokeo's B2B product expansion, developing customer-facing functionality and secure backend data workflows aligned with compliance standards.
- Maintained and extended the company's internal CMS for customer support using Sanity and TypeScript, ensuring seamless data management and user experience.
- Researched and introduced emerging technologies such as AWS, Cypress, and Next.js, accelerating product delivery and improving testing automation.
- Partnered with Engineering, Product, and Executive leadership to prioritize roadmap initiatives and deliver high-quality, business-critical features on schedule.
- Skills: Python, JavaScript, Typescript, Go, FastAPI, React.js, Next.js, Node.js, Pandas, PyTorch, AWS, SQL, PostgreSQL, LangChain, AI/LLM/NLP, Redis, GraphQL, CMS, Docker, Cypress, TailwindCSS, Sanity

Senior Software Engineer

May 2014 – Jul 2020

OpenX Technologies

- Designed and developed multiple single-page applications (SPAs) to power OpenX's supply-side advertising platform, optimizing ad delivery performance across screens using React.js, Python, GCP, and Docker.
- Rewrote and optimized core API services to overcome scalability limitations, enabling the system to process billions of ad requests daily and efficiently connect thousands of publishers with demand partners via Python, MySQL, Java, and RabbitMQ.
- Revamped the company's bidding engine in Golang, boosting throughput from 200 to 10,000 queries per second per server — a 50x performance increase that reduced hardware costs by \$25 million.
- Re-architected prebid reporting ingestion pipelines into Google BigQuery, achieving a 10x reduction in operational costs, and migrated performance metrics and tracing data to Prometheus for improved observability.
- Consolidated and refactored the core ad delivery system built in JavaScript, Python, and Perl, simplifying architecture, reducing latency, and improving system resilience.
- Prototyped the first FLEDGE-based SSP integration through PrebidJS and Google Ad Manager (GAM), optimizing DSP bid times and increasing annual revenue by 15%.
- Collaborated in an agile SCRUM environment alongside full-stack, AI/ML, data science, and DevOps teams, fostering cross-functional innovation and accelerated delivery of new features.
- Skills: React.js, JavaScript, TypeScript, Python, Go (Golang), Java, Perl, RabbitMQ, MySQL, PostgreSQL, GCP, Docker, Kubernetes, BigQuery, Prometheus, PrebidJS, Google Ad Manager (GAM), FLEDGE, Microservices, Distributed Systems, AI/ML Integrations, Agile/SCRUM

Software Engineer

Mar 2011 - May 2014

Yahoo

- Served as a Python backend engineer on Yahoo's Content Analysis Platform (CAP) team, designing and maintaining large-scale systems for data analytics, automation, and content intelligence using Django.
- Architected and implemented end-to-end machine learning pipelines for predictive modeling and data automation, leveraging Hadoop, Mahout, and MongoDB to process and analyze high-volume content data.
- Built a Python-based utility framework using SciPy, NumPy, and Pandas to optimize large-scale data ingestion, feature extraction, and statistical analysis, improving data pipeline efficiency by over 30%.
- Developed and optimized SQL-based ETL pipelines, including schema design, complex queries, and relational data modeling to support advanced content analytics and reporting workflows.
- Integrated AWS CloudSearch to enhance NLP-driven document classification and content retrieval, dynamically labeling and managing training datasets through automated Python scripts.
- Created custom email-tracking and campaign analytics tools using Python-based ETL pipelines, improving targeting precision and boosting marketing engagement rates.
- Led velocity-based release planning and sprint coordination, managing feature prioritization, cross-team dependencies, and customer-facing deliverables to ensure timely and high-quality releases.
- Skills: Python, Django, SciPy, NumPy, Pandas, Hadoop, Mahout, MongoDB, SQL, ETL, AWS CloudSearch, NLP, Data Analytics, Predictive Modeling, Machine Learning Pipelines

EDUCATION

Master's Degree in Computer Science | 2008 ~ 2011

Stanford University

Bachelor's Degree in Electrical Engineering | 2005 ~ 2008

Stanford University

SKILLS

- **Languages:** Python, JavaScript, TypeScript, Go(Golang), Java, SQL, Perl, GraphQL
- **Frameworks:** Django, FastAPI, React.js, Next.js, Node.js, Pandas, NumPy, PyTorch, LangChain, D3.js
- **Databases:** PostgreSQL, MySQL, MongoDB, AWS Dynamo DB, Oracle, Redis
- **Cloud & Infrastructure:** AWS, GCP, Azure, Docker, Kubernetes, OpenSearch, BigQuery, Prometheus
- **AI/ML & Data Technologies:** AI/LLM/NLP Integration, Data Analytics, ETL Pipelines, OpenAI API, Gemini API, Airflow, Tableau